



Lecture Session 11

1. Forces acting on the ocean;
2. Earth Rotation: The Coriolis effect;
3. Driving mechanisms of ocean currents.

Icebreaker discussion

- What are forces?
- How do forces act on objects? (Hint: Remember Newton)
- Based on the video, list the forces you think are acting on the ocean.

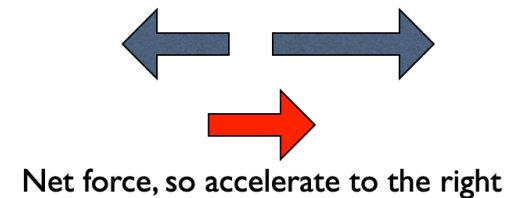


1. What are forces?

Like energy, forces are extremely hard to define. In summary, a force is an influence that can change the motion of an object (solid or fluid).

2. How do forces act on objects? (Hint: Remember Newton)

Both ocean and atmosphere physical behaviors can be studied using the good old Newton's second law: $[Sum\ of\ Forces] = mass \times acceleration$



3. Based on the video, list the forces you think are acting on the ocean.

There are mainly 3 forces acting on the ocean (and atmosphere):

1. The pressure gradient;
2. Friction;
3. Gravity (buoyancy)

Especial effect: The Coriolis effect (apparent force due to Earth's rotation).

An aerial photograph of a coral reef system. The top left shows deep blue ocean water. A curved reef edge separates this from a shallower area with green and brownish water, indicating coral and sand. Further out, the water is a lighter greenish-blue. The text 'The pressure gradient force' is centered in white.

The pressure gradient force

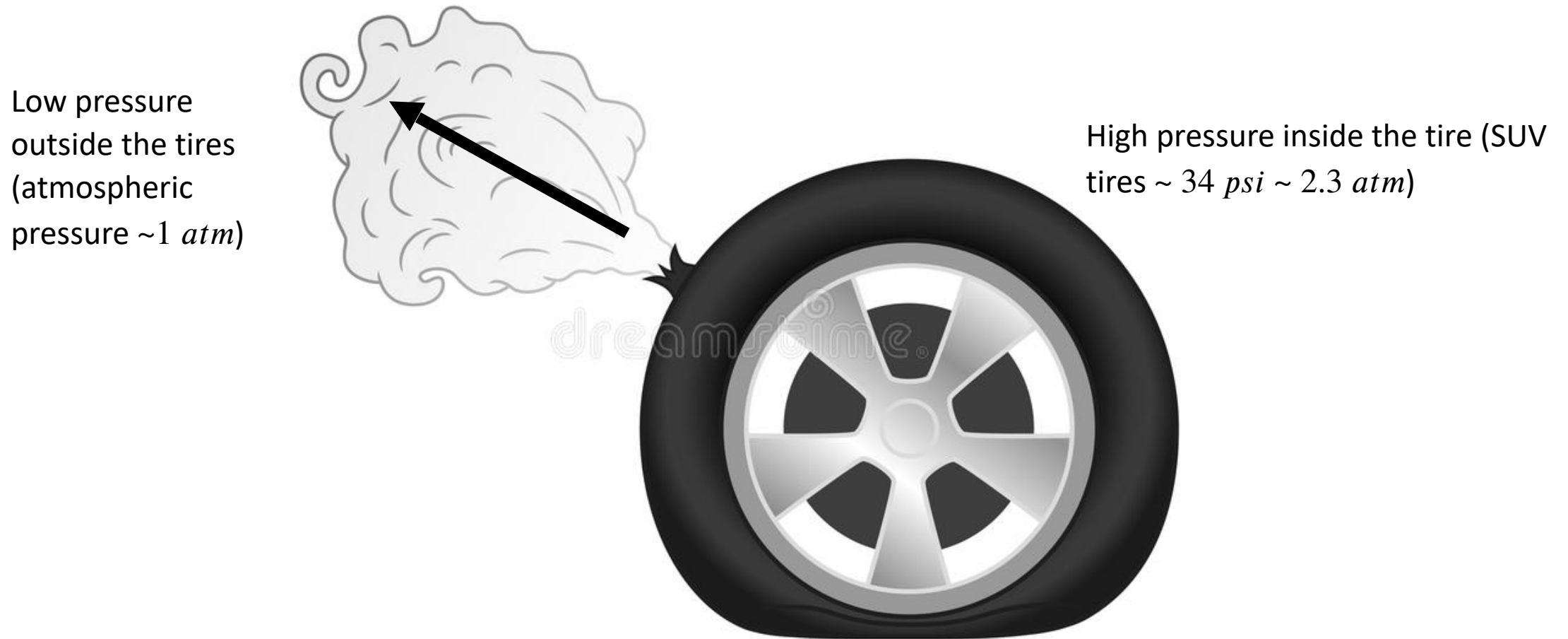
The Pressure Gradient

Air moving out of a tire



The Pressure Gradient

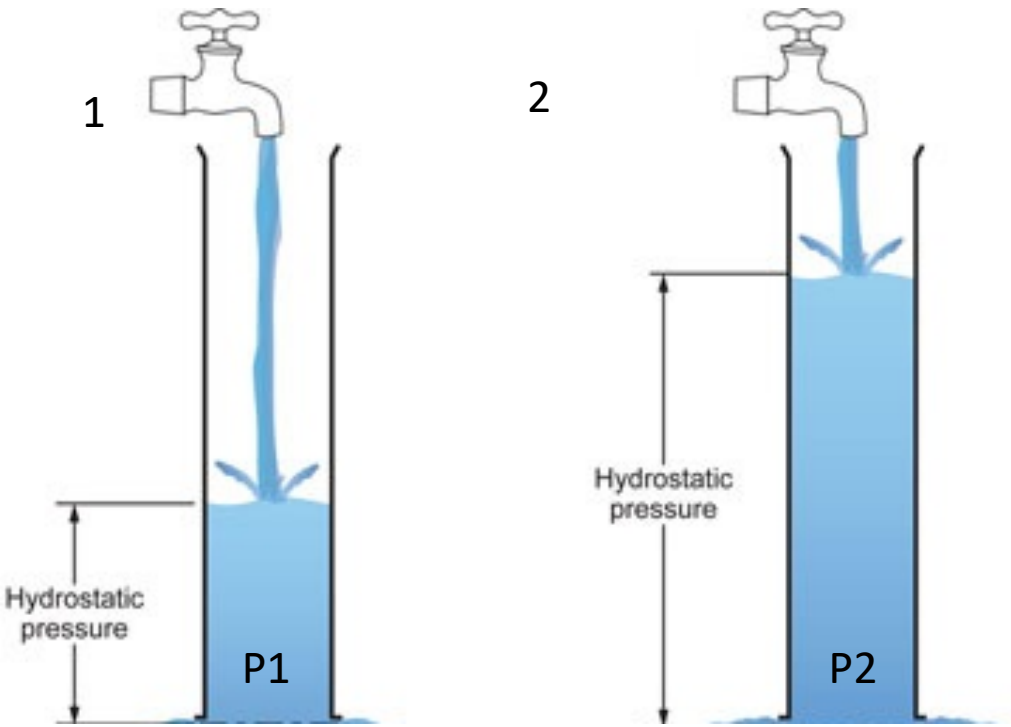
Fluids will tend to accelerate in the presence of a pressure gradient. Difference in pressure generates a force that pushes fluids toward lower pressures. Larger the gradient (or pressure difference) large will be the force.



The Pressure Gradient

Another example

P = pressure at the bottom of each tube

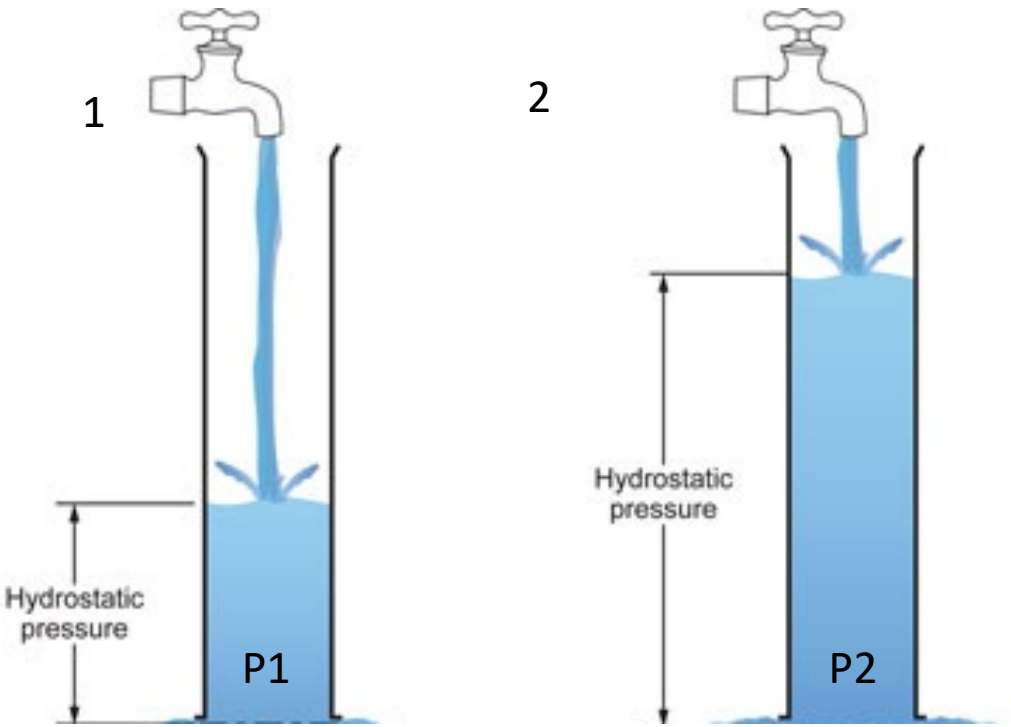


More water in tube 2 results in $P_2 > P_1$

The Pressure Gradient

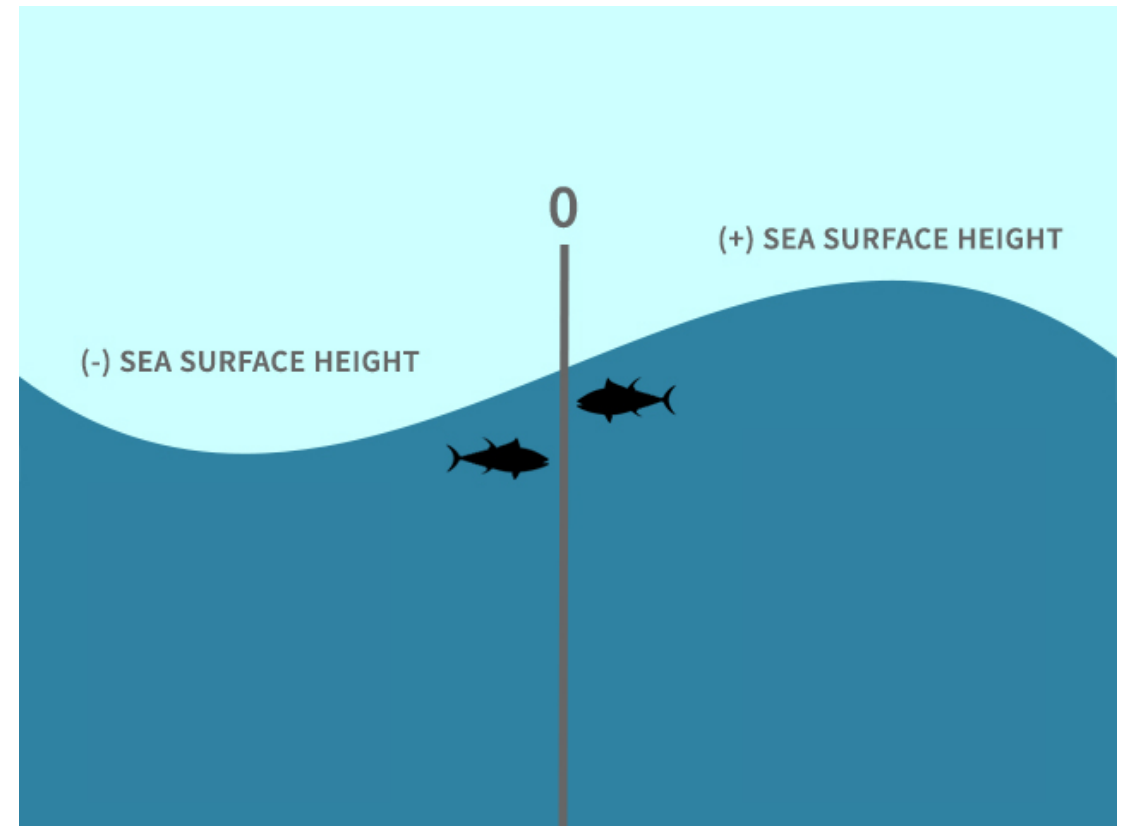
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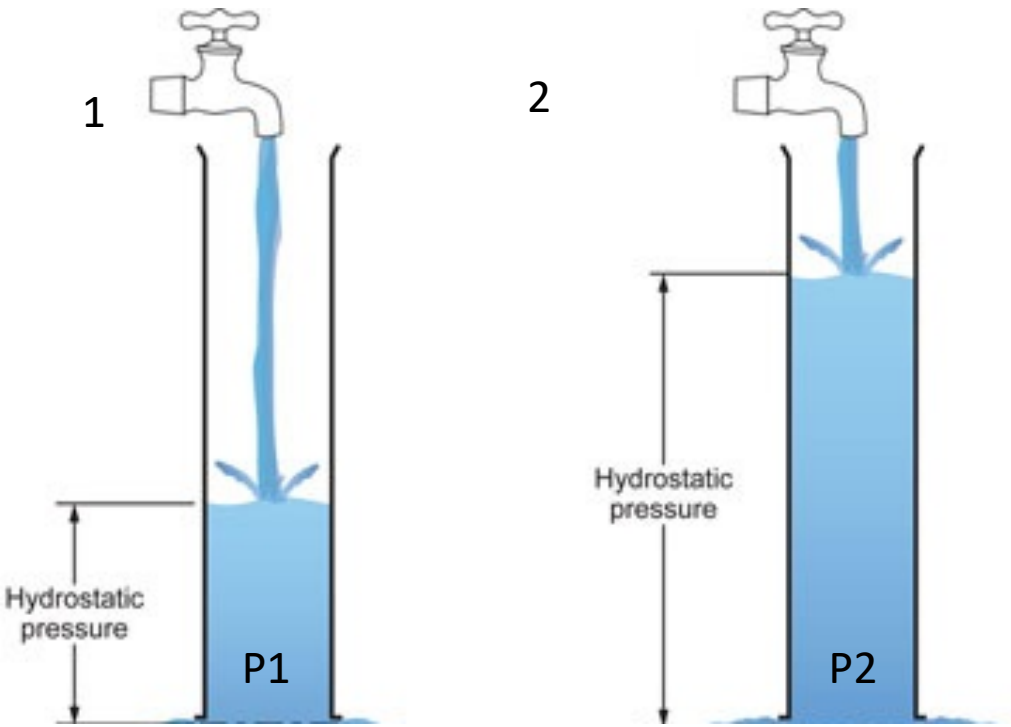
In the Ocean: Where will water tend to flow?



The Pressure Gradient

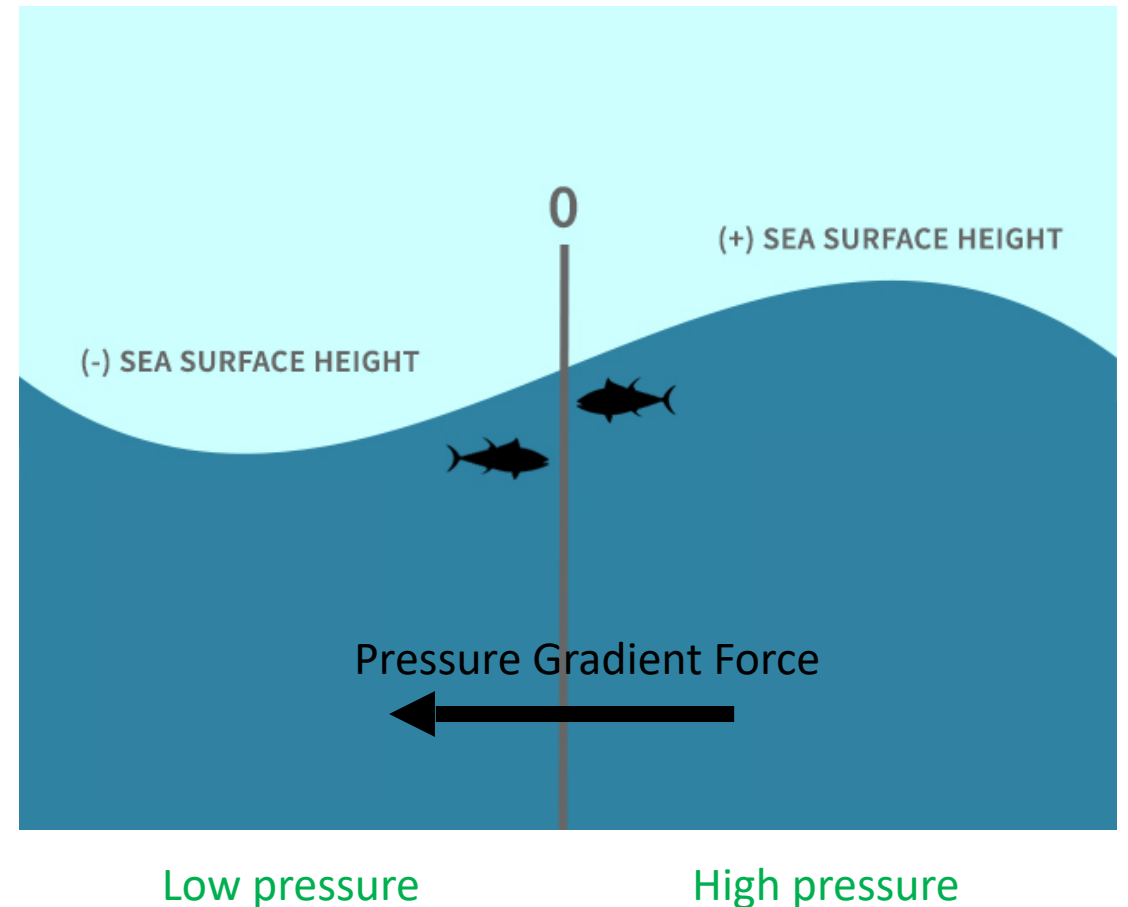
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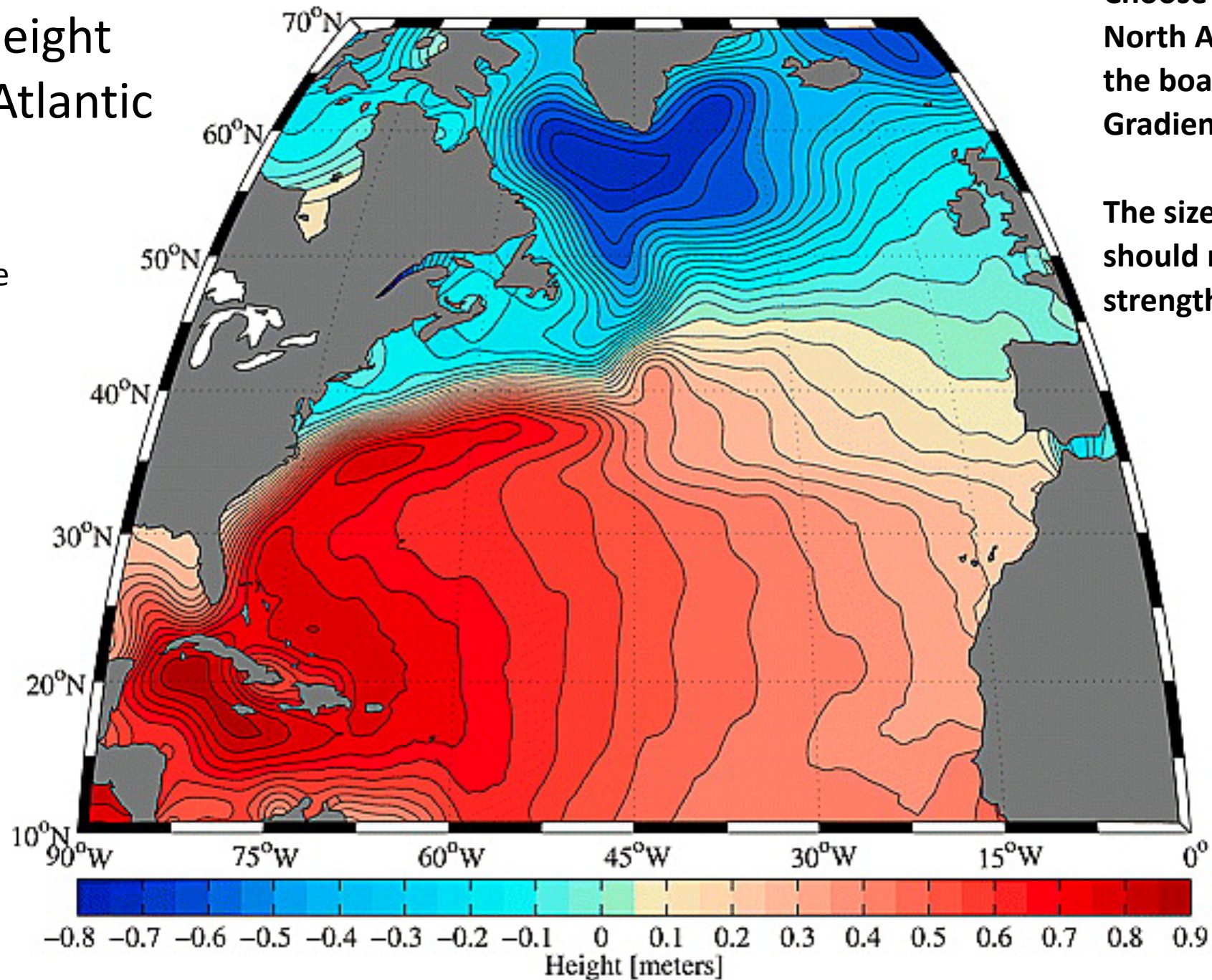
In the Ocean: Where will water tend to flow?



Sea Surface Height of the North Atlantic

Negative: Sea Surface
Depressions

Positive: Sea Surface
Hills



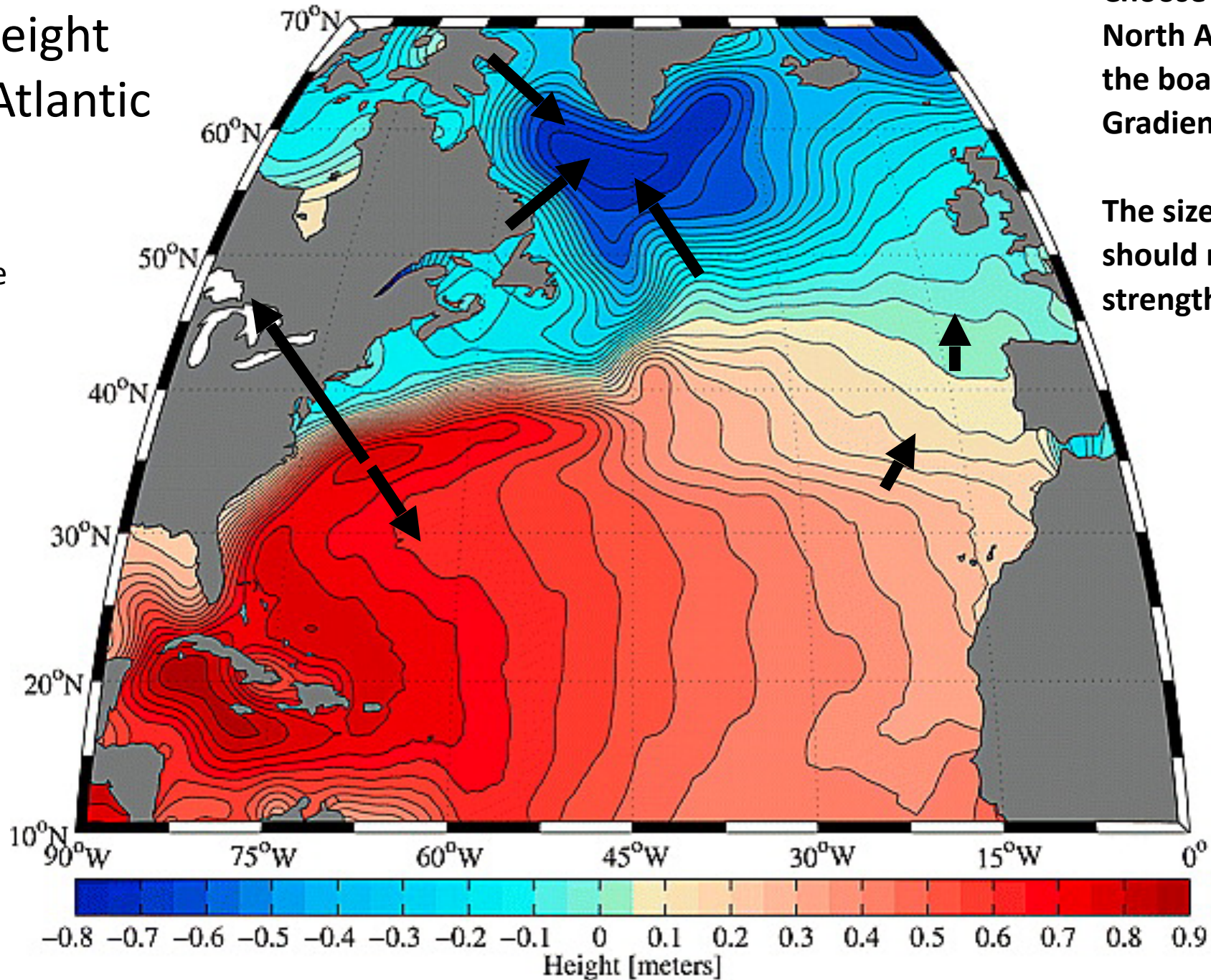
Choose an area of the
North Atlantic and draw on
the board the Pressure
Gradient Force vector.

The size of the vector
should reflect the size
strength of the force.

Sea Surface Height of the North Atlantic

Negative: Sea Surface Depressions

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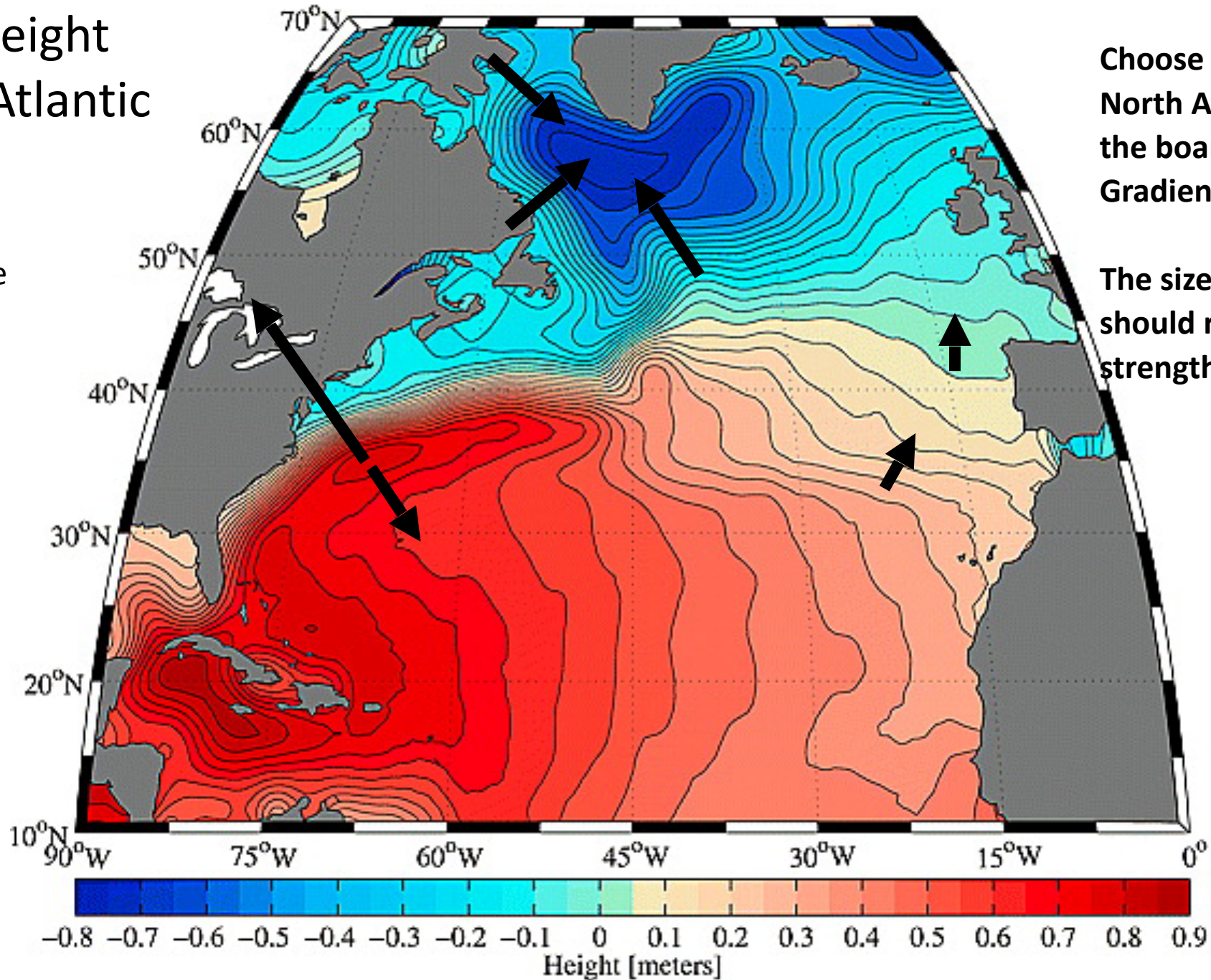
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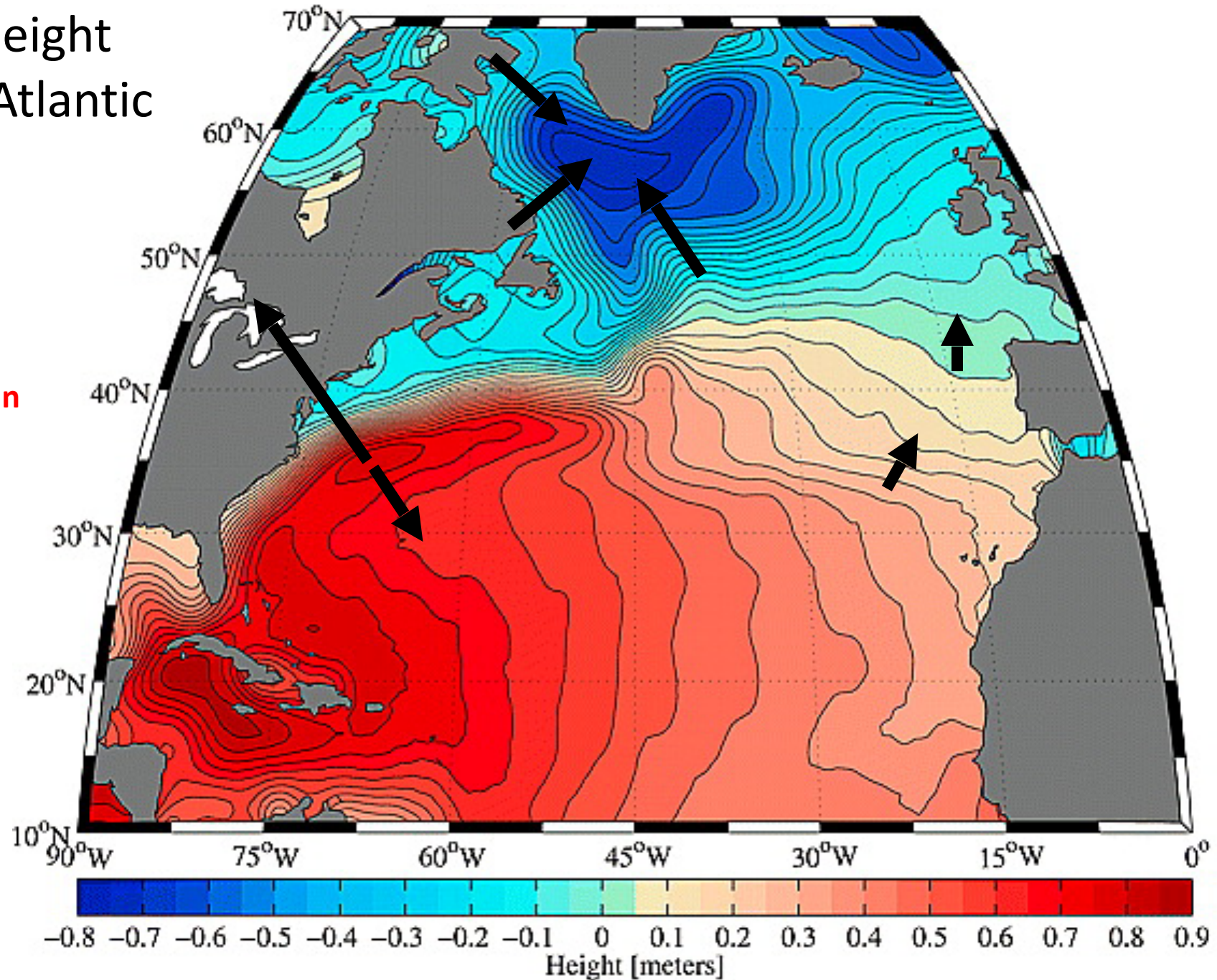
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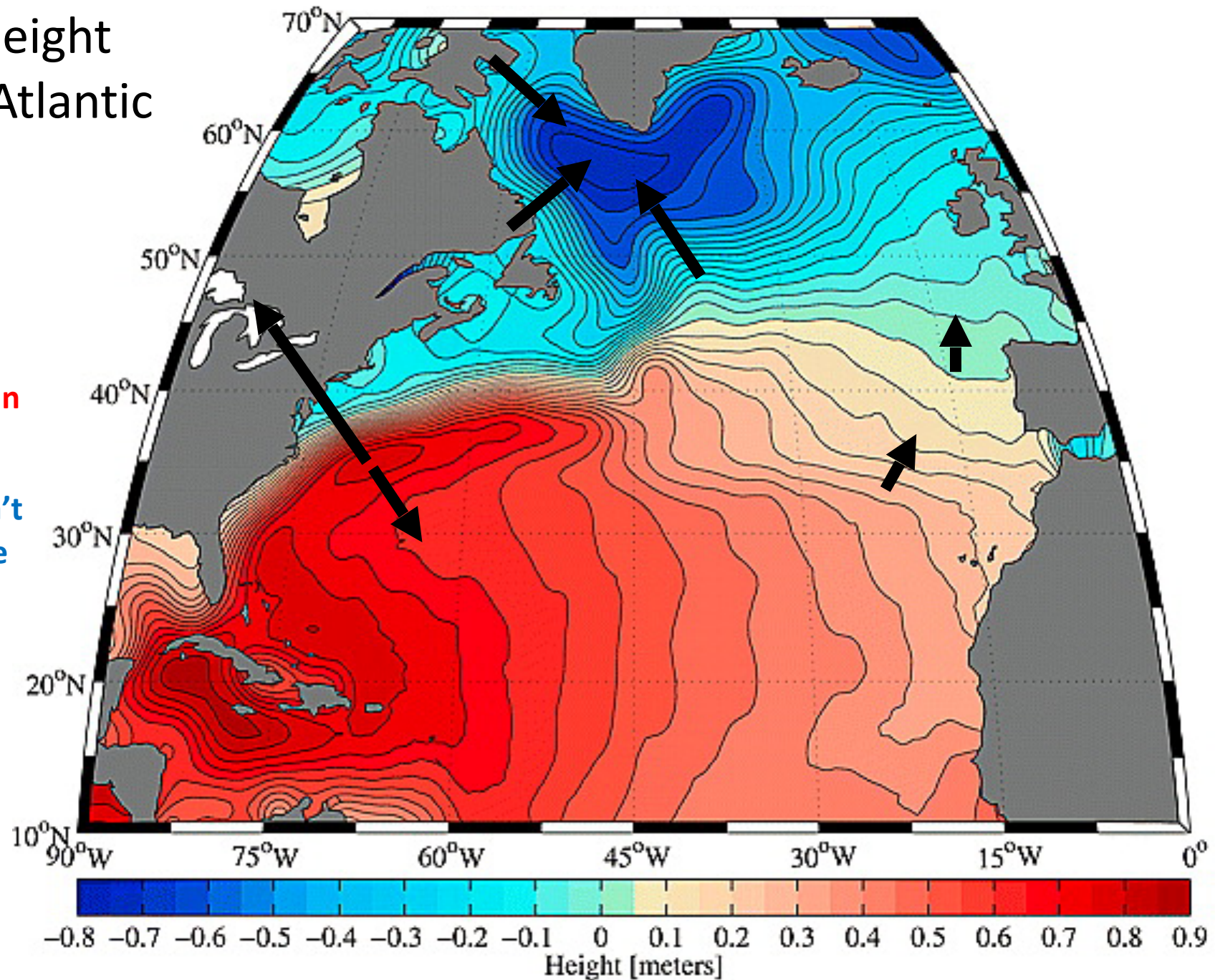
Do you think that is direction of the ocean currents? Why?



Sea Surface Height of the North Atlantic

Do you think that is direction of the ocean currents? Why?

No, because we didn't take into account the other forces.

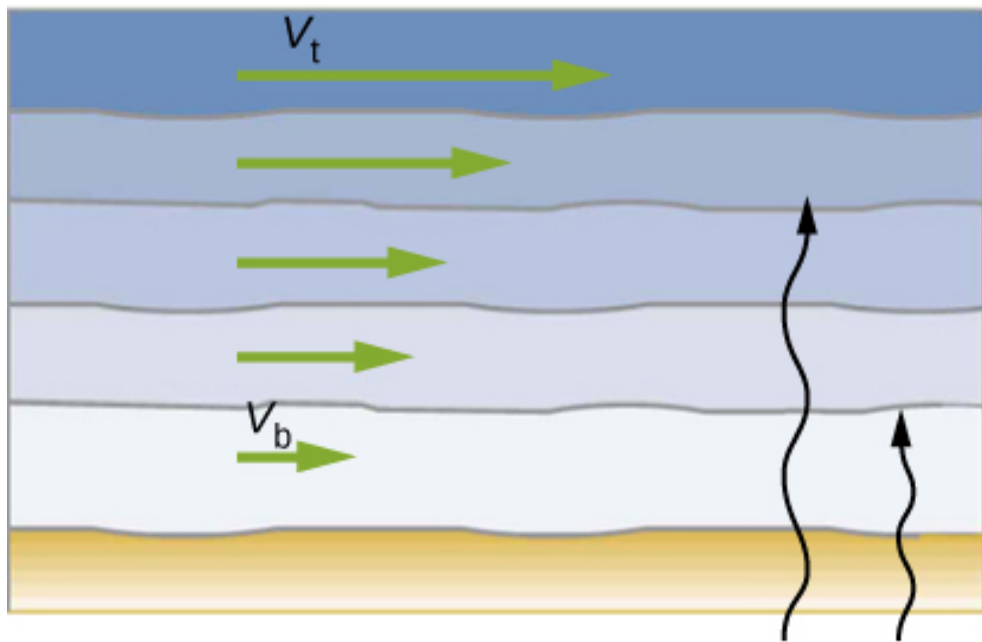


An aerial photograph of a coral reef system. The top left shows deep blue ocean water. A diagonal line of coral reef separates the deep water from a shallower area. The reef has a complex, textured appearance with various shades of green and brown. To the right of the reef, there is a large, light-colored, irregularly shaped area that appears to be a lagoon or a sandy beach. The text "Frictional forces" is overlaid in white, sans-serif font across the middle of the image, positioned over the reef and the lagoon area.

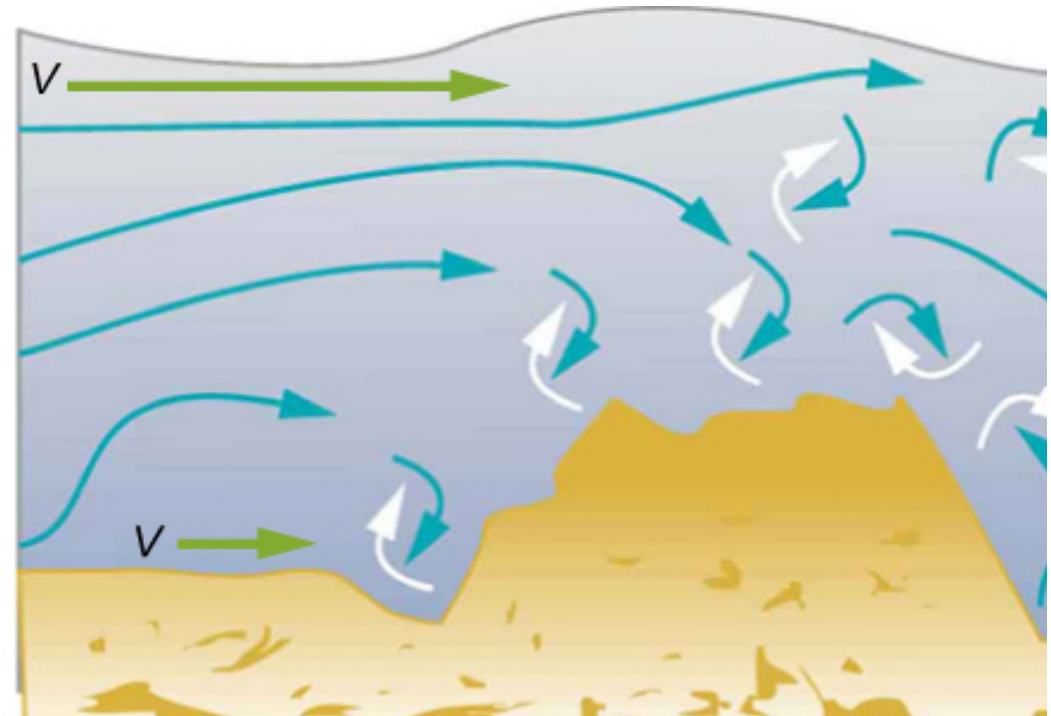
Frictional forces

Friction

Although water can reduce friction between surfaces (like a lubricant), seawater and the atmosphere are not completely inviscid. Therefore, water parcels traveling at different speeds can rub against each other due to friction, modifying the ocean currents. The friction between the ocean and atmosphere (winds) generate the winds and accelerate the ocean currents near the surface. In contrast, friction with the bottom slow down the ocean currents.



Friction between layers



Friction

Although water can reduce friction between surfaces (like a lubricant), seawater and the atmosphere are not completely inviscid. Therefore, water parcels traveling at different speeds can rub against each other due to friction, modifying the ocean currents. The friction between the ocean and atmosphere (winds) generate waves and accelerate the ocean currents near the surface. In contrast, friction with the bottom slow down the ocean currents.

Strong winds of a storm generating large waves



An aerial photograph of a tropical coastline. A narrow white sand beach runs diagonally across the frame, separating a shallow turquoise lagoon from a deeper, dark blue ocean. The lagoon floor is visible through the clear water, showing sandy and coral patches. In the upper right, there are some white, rocky or coral formations. The text "Gravitational force" is centered over the image in a white, sans-serif font.

Gravitational force

Gravity

Force that attracts any two objects with mass. In fluids like the ocean and atmosphere, the Earth's gravity is opposed by the buoyancy of water, generating a stable vertical layered stack of fluid sorted by the density of each layer (lighter layers resting on top of heavier layer).

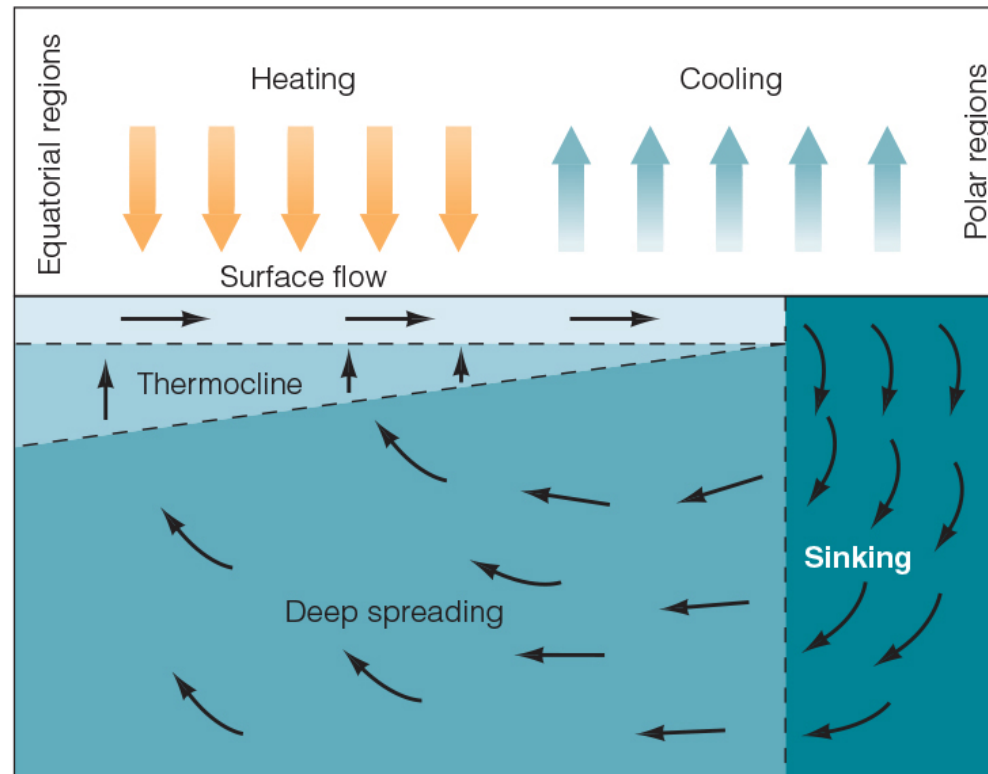
However, external factors can disrupt this stable configuration, allowing gravity (buoyancy) to accelerate the flow in the vertical (Example: Convective currents discussed in previous class).

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The Thermohaline circulation is an example of gravity (or buoyancy) driven circulation.

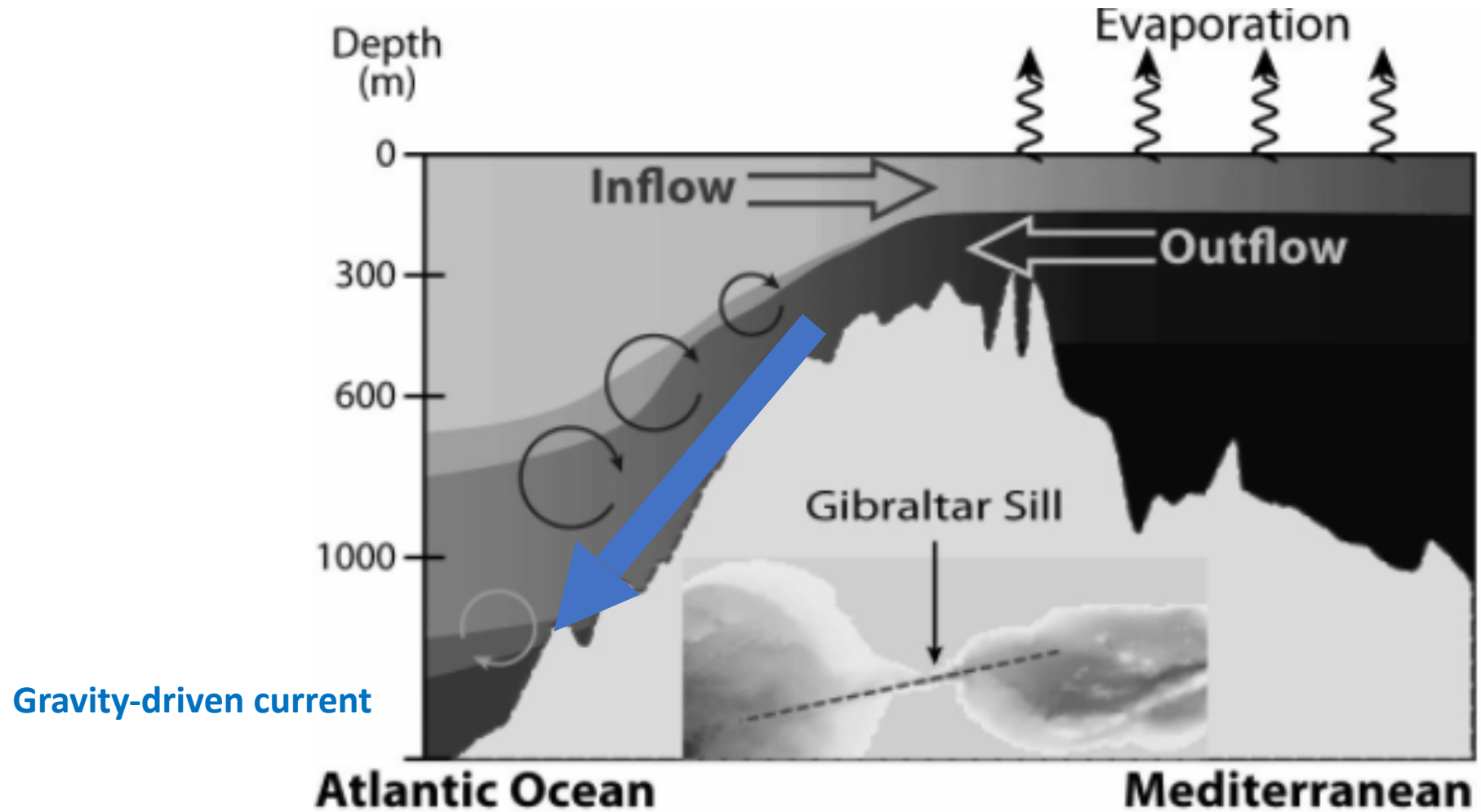


Evaporation and precipitation over the ocean's surface can have the similar effect of cooling and heating, respectively. Salinity will also affect density.

Figure 9.34 The classic model of a pure thermohaline circulation, caused by heating in lower latitudes and cooling in higher latitudes. Compare this figure to the convection cell shown in Figure 8.9.

Gravity

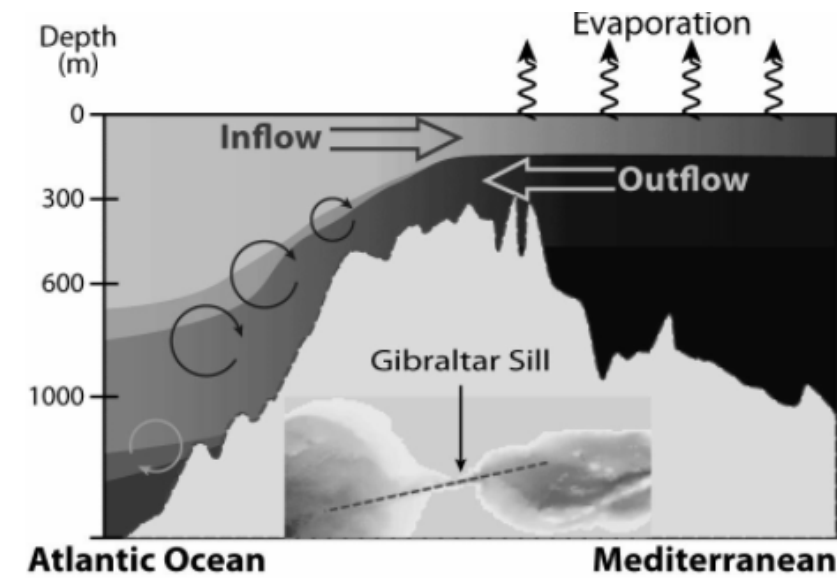
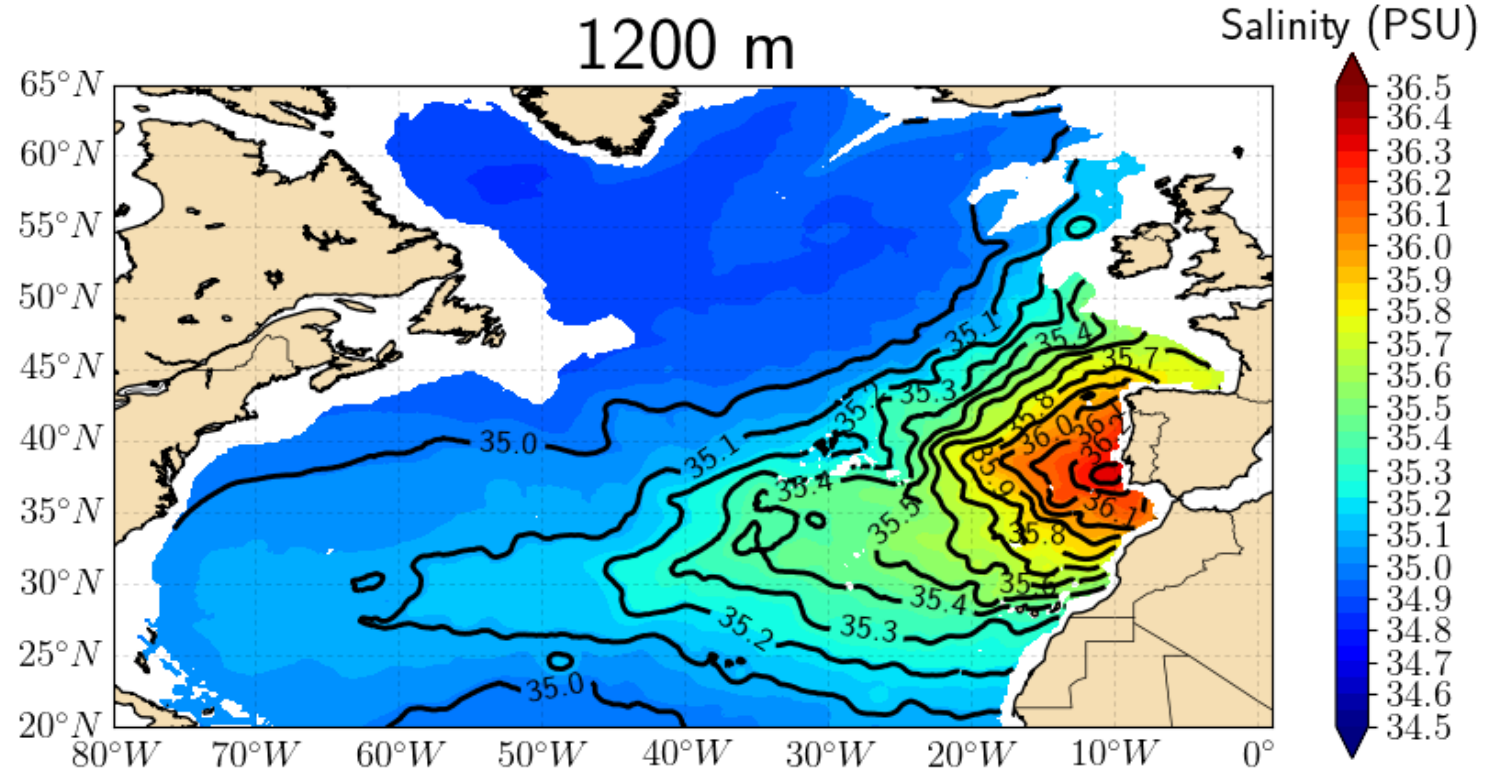
The Mediterranean Overflow Waters



Gravity

The Mediterranean Overflow Waters

Salinity at 1200 m: The Mediterranean Salt Tongue



An aerial photograph of a tropical coastline. The top half of the image shows deep blue ocean water. A narrow white sandy beach runs diagonally across the middle. Below the beach is a shallow lagoon with light green and yellowish water, showing some darker patches of vegetation or sand. The bottom right corner shows more of the lagoon and some white sand. The text 'The Coriolis effect' is overlaid in white, centered horizontally.

The Coriolis effect

Things are about to get weird

The Coriolis Effect

Apparent force that arises from the fact the Earth is rotating from west to east around its own axis. It deflects objects moving slow or long enough (large distances) to the right in the northern hemisphere, or to the left in the southern hemisphere.

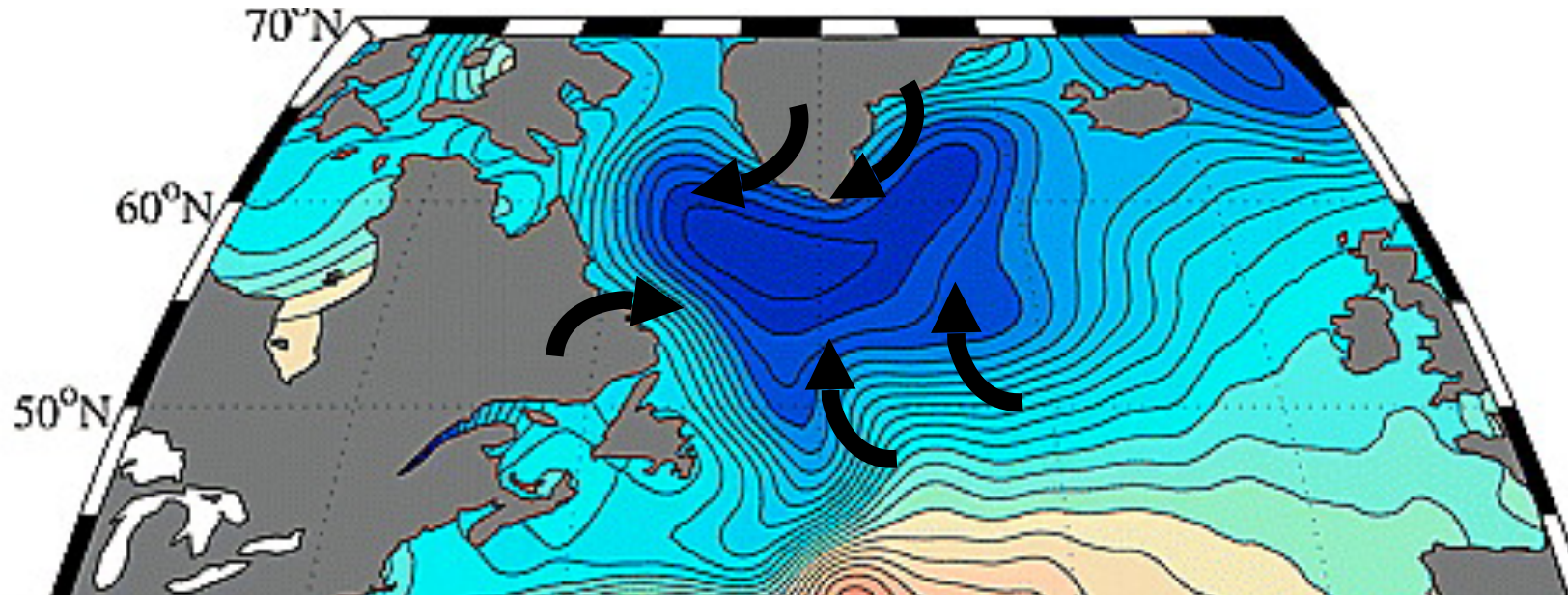


It is not an illusion. It is a real effect!

The Coriolis Effect

The Subpolar North Atlantic counter-clockwise circulation example. The pressure gradient force will tend to push waters towards the low-pressure region. However, the Coriolis force will always pull the moving currents to the right counterbalancing the pressure gradient.

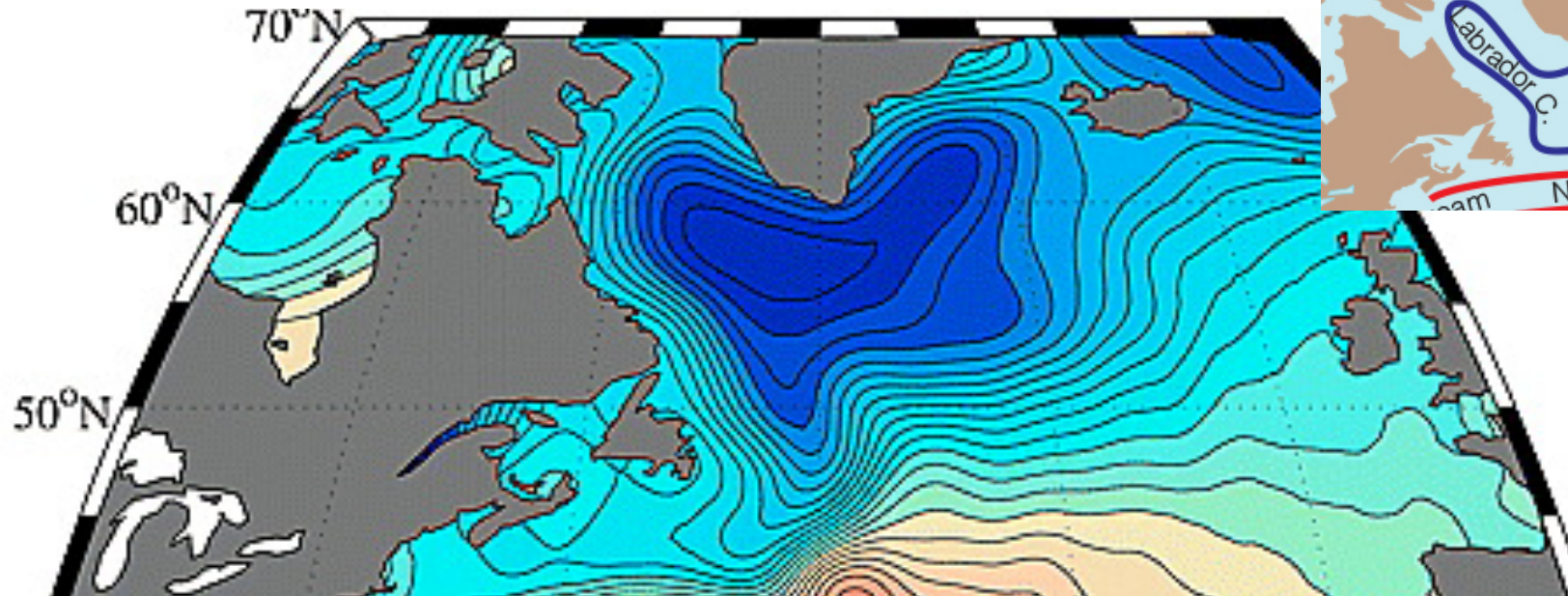
Sea Surface Height: Blue means sea surface depressions or low pressure



The Coriolis Effect

The Subpolar North Atlantic counter-clockwise circulation example. The pressure gradient force will tend to push waters towards the low-pressure region. However, the Coriolis force will always pull the moving currents to the right counterbalancing the pressure gradient.

Sea Surface Height: Blue means sea surface depressions or low pressure



As a result, a closed counter-clockwise circulation will form around the sea surface depression.